

Naghmeh (Melody) Nazar

AFFECTIVE COMPUTING | HEALTH AI | HUMAN-CENTERED MACHINE LEARNING

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

RESEARCH PROFILE

Machine-learning researcher with an M.S. in Computer Engineering from California State University, Northridge, studying real-time speech emotion recognition. Work combines audio signal processing, deep residual networks, speaker-aware evaluation, and deployable inference. Interested in robust multimodal sensing, affective computing, and responsible health AI, with careful attention to domain shift, subgroup performance, and reproducibility.

EDUCATION

M.S. in Computer Engineering, California State University, Northridge

Aug 2024 - Aug 2026

- GPA: 3.5/4.0
- Thesis: Real-Time Speech Emotion Recognition Using Optimized Deep Residual Networks (ResNet)
- Advisor: Prof. Shahnam Mirzaei

B.S. in Computer Engineering, University of Science and Technology of Mazandaran

Aug 2016 - Mar 2021

MANUSCRIPT & PRESENTATION

- IEEE SLT 2026 submission based on the M.S. thesis; manuscript under review.
- Presented the speech emotion recognition research at the 40th Annual CSU Student Research Competition.

RESEARCH EXPERIENCE

Real-Time Speech Emotion Recognition - M.S. Thesis, CSUN

2024 - 2026

- Developed an end-to-end four-class system for neutral, happy, sad, and angry speech using a 2D deep residual network.
- Standardized approximately 4,900 RAVDESS and TESS utterances at 16 kHz and generated 128 x 128 Mel-spectrograms with a speaker-aware train/validation/test partition.
- Reached 80.36% peak validation accuracy and 74.4% held-out mixed-test accuracy; analyzed class errors, Happy/Neutral ambiguity, speaker variability, and cross-corpus shift.
- Stabilized training with Adam, gradient clipping (clip norm 1.0), learning-rate reduction from 1e-3 to 1e-7, early stopping, and reproducible preprocessing.
- Implemented a near-real-time desktop CPU prototype and identified cross-corpus shift and speaker variability as central barriers to naturalistic and health-related use; no clinical validation is claimed.

Persian COVID-19 Social Behavior Analysis - NLP Research

2021

- Analyzed Persian-language social-media posts using sentiment classification, topic modeling, and temporal trend analysis.

SELECTED PROJECTS

CCTV Product-Support Chatbot

Python, Dialogflow, Google Cloud

- Designed an intent-driven assistant that guides users through CCTV product selection and support questions.
- Integrated Dialogflow NLP with cloud-backed services and structured conversational flows for real-time interaction.

GenAI Email & Scheduling Agent

n8n, Gmail, Airtable, Google Calendar, structured JSON

- Built a tool-integrated workflow that classifies incoming email, retrieves business context, and proposes scheduling actions based on user intent and availability.
- Used structured outputs, API integrations, approval logic, and prompt refinement to reduce redundant tool calls and make workflow behavior easier to inspect.

PROFESSIONAL EXPERIENCE

Senior Frontend Developer, Golrang Industrial Group - Tehran, Iran

Jul 2022 - Jul 2024

- Developed and maintained production web and mobile applications using React, React Native, Vue.js, TypeScript, and modern state-management patterns.
- Improved responsiveness and maintainability through profiling, caching, code splitting, debugging, and systematic code review.

React Developer, Erole.ir - Tehran, Iran

Jun 2021 - Jun 2022

- Built responsive React/Redux applications and integrated REST APIs and SignalR-based real-time features.

TECHNICAL SKILLS

- **Machine Learning:** Python, TensorFlow, Keras, scikit-learn, NumPy, Pandas, CNNs, ResNet, GRU/RNN, classification, evaluation
- **Signal & Behavioral Data:** audio preprocessing, Mel-spectrograms, MFCCs, time-frequency analysis, affective computing, NLP, topic modeling
- **Research Tools:** SciPy, Librosa, Jupyter, Google Colab, Git, Linux, reproducible experiment workflows
- **Engineering:** C++, JavaScript, TypeScript, React, Vue.js, Node.js, REST APIs